

**Fly your bike**

If you own a Honda Reflex, buy the Flite Bike kit and watch as your wheels take to the air!

**Air powered**

Indian Engineering students have made a 100 cc bike that runs on compressed air

**ECU Flashing**

The expensive and often dangerous "hobby" of changing the firmware in ECUs for custom tuning

An insight into motorcycle gadgetry

Wonder Wheels



Photos: YAMAHA RACING, HONDA

Joshua Crasto

Modern boy racers in the premier class of motorcycle racing have been debating the 'excessive' use of electronics that currently subdue their 250-odd bhp motorcycles. Some are of the opinion that it is progressively killing the sport, others think they're simply a way of making the motorcycling world a better and safer place to ride in. The men who made their mark in the racing world in the 80s and 90s were unanimous in saying that electronics have their time and place – Monday to Friday, 9 to 5, at the R&D centre. But then again, if it means living life the way they'd like it, we'd all show up to work boozed up, black-eyed and have a

leggy blonde for a personal secretary...

The blonde secretary is a myth and it has been 12 hours since you had your last beer (if

have over the years trickled down to the everyday road bike. So let's spend some time going over a motorcycle to closely examine these systems

Let's grab a microscope and put the key into the ignition of a 2008 Yamaha YZF-R6 and see what happens. The tach needle sweeps through its range while that little black box silently boots up: sensors feed signals to the ECU, kickstand up...

you're lucky) and we're now in the 21st century. This is the age of little black boxes, tiny microchips and a host of other silicon-smothered devices. Developed for state-of-the-art race motorcycles, these gadgets

and see how, and if, they make a difference to your life.

The think tank

The Electronic Control Unit

Let us start by examining the core of every current

day motorcycle. Like your first guess, we too would like to believe this to be the crankshaft, the pistons or at least the motor in its entirety, but the truth is far from it. It's a lunch-box sized "brain" called the ECU. A small black box that mad scientists in white lab coats decided would have a 32-bit CPU running at about 40 MHz. The numbers pale in comparison to your average desktop, but with miniscule, clever code, the CPU is a thousand times more efficient. The ECU micro-manages every little happening on the motorcycle, working in collaboration with numerous sensors from systems such as the lighting and the brakes, to the fuel and air flowing into and out of the engine. And these are just its primary functions. Let's